

H.SI ORGANIC PART:

NEW: [From hostel Eval]

① Rxn to show acidic nature of H_2SO_4 ,1) Diff between bleaching action of SO_2 and Cl_2 [3]

2) What is homologous series? Write characteristics of homologous series. [2]

3) What happens when,

i) conc. H_2SO_4 is treated with glucose?ii) SO_2 is passed over moist red flower?

b) Define functional groups with examples. [1]

c) What is vital force theory? Why rejected? [1]

1) H_2SO_4 is known as king of chemical why?a) Flowsheet Diagram for prep of H_2SO_4 by contact [2]b) Conditions for optimum yield of H_2SO_4 . [2]② Ch reaction to illustrate reducing property of SO_2 . [1] + H_2S :

d) Define tetra covalency and catenation of carbon.

e) Define empirical formula and bond line form. [with eg: [2]

f) Reasons for studying organic comp as separate branch of chemistry.

MCA: Cetane no of diesel fuel can be increased by the add of:

③ Sodium extract must be boiled with HNO_3 before test

LONG: Define octane no with eg

Define gasoline additive with eg

Why sodium extract is generally alkaline in nature?

d) How to detect Sulphur & halogen as hetero element? Explain with necessary reaction

OLD :

- 1) IONIZATION ENERGY / ELECTRON AFFINITY
± Vary with group and period: (2)
- 2) Diff Electron affinity and electronegativity.
- 3) Bohr's classification / Diff electron classification.
- 4) Ionic size comparison:
eg: F^- , Ne & Na^+
- 5) Short note on homologous series & character.
- 6) Diff between organic and inorganic:
- 7) Kipps apparatus +
- 8) H_2SO_4 as oxidizing / dehydrating / acid / precipitating agent.

H_2S reducing nature, precipitating agent

Affinity of Be/Mg Zero, (1) F

PERIODIC TABLE:

- i) Screening effect
- ii) Van der Waals radius + other,
- iii) isoelectronic species + compare size of Cl^- , Ar , &
HETEROCYCLIC COMPOUND + IUPAC Names.